

Name: _____



Problem solving with factors, multiples, squares and cubes

Task A: Solve problems with squares and cubes

- 1)
 - a) What is the largest square you can make with 80 blocks?

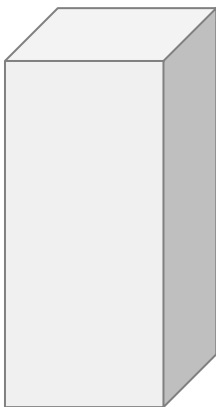
 - b) What is the largest square you can make with 200 blocks?

 - c) What is the largest cube you can make with 200 blocks?

 - d) What is the largest cube you can make with 350 blocks?

 - e) Justify why 8 is not a square number.

- 2) The volume of a cuboid is 90 cm^3 .
If the dimensions are all integers, what could they be?



Task B: Solve problems with factors and multiples

- 1) Each of the following rows contains consecutive numbers which satisfy the condition given. No number is greater than 20.

a)

three factors	sum of digits is 1	odd	multiple of 3
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b)

sum of digits is 5	sum of digits is 6	five factors	prime
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c)

four factors	prime	four factors	three factors
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- 2) Use the product of prime factors to complete these grids.

a)

			154
			30
			20
20	70	66	

$$154 = 2 \times 7 \times 11$$

$$30 = 2 \times 3 \times 5$$

$$20 = 2 \times 2 \times 5$$

$$70 = 2 \times 5 \times 7$$

$$66 = 2 \times 3 \times 11$$

- b) Use your calculator to write these numbers as a product of prime factors to complete the grid

			275
			99
			70
105	110	165	